Rajwinder Singh

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Education

The University of Sydney — Sydney, New South Wales
Master of Professional Engineering, Aerospace. — Expected Graduation: 2028
Recipient of the Sydney International Student Award

Punjab Engineering College — Chandigarh, India
Bachelor of Technology, Aerospace — 2021 – 2025
CGPA - 9.09/10.00
Recipient of the best Capstone Project Award in Aerospace Discipline, 2025

Delhi Public School — Chandigarh, India
CBSE Class 12 Result - 93% — 2020

Holy Family Convent School — Ropar, Punjab, India
ICSE Class 10 Result - 95% — 2018

Experience

ARTPARK, Indian Institute of Science (IISc) — Bengaluru, Karnataka, India
Testing and Instrumentation Engineer (Intern) — January 2024 – June 2024

- Key contributions in UAV propulsion R&D, which include first principles modelling and experimental test setup and validation of propulsion systems.
- Developed a Blade-Element framework in MATLAB to model and analyze UAV propellers.
- Developed a propulsion data-handling software in Python to post-process large time-stamped datasets, reducing processing times by 50%.
- Developed data-driven semi-analytic methods in MATLAB, to characterize the heat-transfer behaviour of propulsion systems during in-flight operations.
- Contributed to the assembly and firmware development in embedded C/C++, of custom data acquisition setup tailored for propulsion systems.
- Routine sensor integration, test-rig maintenance and troubleshooting of propulsion systems

Projects

Aerodynamic Design of A Formula 1 Racing Vehicle — Chandigarh, India
Team Leader — August 2024 – August 2025

- Capstone project (Best Capstone Project Award in Aerospace Discipline, 2025). Led end-to-end aerodynamic design, CFD simulation, and optimization workflow for an F1 vehicle..
- Designed, evaluated, and optimized multiple F1 vehicle concepts in compliance with 2026 FIA regulations.
- Conducted CFD analyses using OpenFOAM; CAD and CAM modeling with Autodesk Fusion 360.
- Developed a C++/Qt physics-based software for RANS parameter tuning, boundary/initial conditions, and mesh refinement control in OpenFOAM.
- Implemented an aerodynamic shape optimizer accelerated by a neural-network surrogate model in PyTorch for aerodynamic evaluation.
- Fabricated a 1:10 scale 3D printed prototype of the final design for wind tunnel testing.

Fast BiQuad Filter!: Real-Time Digital Filter in Modern C++ — Sydney, New South Wales
Team Leader — June 2026 – July 2026

- Developed a real-time VST3 audio plugin in modern C++ implementing low-pass, high-pass, and band-pass biquad filters.
- Designed a low-latency DSP engine performing sample-by-sample filtering with runtime coefficient updates based on cutoff frequency and Q factor.
- Implemented a thread-safe architecture separating the graphical user interface from the real-time audio processing thread.
- Built using iPlug2 and the Steinberg VST3 SDK with emphasis on deterministic real-time execution and modern C++ design.

ferode: Pure Rust Ordinary Differential Equation (ODE) Library — Sydney, New South Wales
June 2025 – August 2025

- ODE solver library written from scratch in Rust for scientific computing applications.
- Designed a memory-safe numerical solver library in pure Rust with reusable abstractions for multiple integration methods.
- Validated five explicit solver schemes, with the most advanced implementation allowing 5th order solution accuracy.
- Validated solver accuracy against standard benchmark problems with adaptive error-controlled stepping.

Oden: Modern C++ Template Library For N-Dimensional ODE Systems — Sydney, New South Wales
June 2026 – Present

- Header-only coupled ODE solver in modern C++23 with emphasis on zero-overhead abstractions and type safety.
- Template-based architecture enabling flexible and extensible solver design for arbitrary systems of equations.
- Implemented and validated baseline adaptive Euler scheme, similar to ferode implementation, but extended to N-dimensional ODE systems.
- Currently designing implementations for adaptive Runge-Kutta schemes, extended to system of ODEs.

Additional Qualifications

Digital Signal Processing (DSP) from the Ground Up in C — Chandigarh, India
2025

- Implemented FIR and IIR filters in C, with applications in areas including avionics.

Introduction To Experiments In Flight, Indian Institute of Technology Kanpur — Kanpur, India
2025

- Measured and analyzed performance, stability, and control parameters of Cessna and Piper aircrafts, in flight.

Leadership and Activities

Sydney Acers Volleyball Club — Sydney, New South Wales
Athlete — February 2026 – Present

- Volleyball Australia registered athlete, representing the club in State-Level Division 1.

Punjab Engineering College Volleyball Team — Chandigarh, India
Captain — 2021 – 2025

- Captain of the national-level college volleyball team. Runner-Up in 2023 Nationals.

Skills & Interests

Programming Languages: C, C++, Rust, Python, MATLAB
Software Engineering: Git, Linux, Modern C++ (C++20/23), Object-Oriented Design
Scientific Computing: Numerical Methods, ODE/PDE Solvers, Computational Fluid Dynamics (CFD), Linear Algebra, Machine Learning (PyTorch), NumPy, Pandas
Engineering Tools: ANSYS, OpenFOAM, SolidWorks, Autodesk Fusion 360
Interests: Music Production, Audio Programming (MaxMSP, PureData), Sound Design